On the prediction of critical transitions



Candidate:
Davide Papapicco

Supervisors:
Graham Donovan, Lauren Smith

Department of Mathematics, Faculty of Science
Waipapa Taumata Rau, The University of Auckland

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Abstract

To be decided.

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Chapter 1

Introduction

The dynamics of complex systems has receveid significant attention, over the last decades, from many research communities in science and engineering. Of particular interest to scientists and modelers is the study of the nonlinear response of these systems to external perturbations which often features the emergence of a rich and diverse set of phenomena including pattern formation, chaotic regimes, resonance and, crucially, critical transitions. A critical transition in a complex system, often referred to as a *tipping point* or a *tipping event*, is recorded whenever a small change in the system's input leads to a large and sometimes irreversible change in the systems' response. More specifically, a tipping point is considered as an abrupt shift in the steady–state of a dynamical system following a slow degradation of the stability of its equilibria.

A concrete example, often discussed in the literaure to introduce the main concepts of critical transitions, is the collapse of multi–species ecosystems under sustained and increasing deterioration of their biomass. In 1977 Robert May proposed a dynamical model in which a population of herbivores, sustained by a decreasing level of vegetation cover in a closed environment, can experience a sudden collapse due to unsustainable increases in the grazing rate [63]. In his model, May realized that a characteristic separation between the timescales of the evolution of the animal population and the grazing rate of the biomass leads to qualitatively different regimes in the observed state of the ecosystem. In one state the ecosystem is in a healthy, sustainable equilibrium and the vegetation cover allows for the slowly increase in the grazing activity of the herbivores. Past a certain threshold however (described as *breakpoint* in May's paper) the ecosystem can no longer sustain the animal population and the biomass sharply declines to near–zero across the environment, resulting in the collapse of the

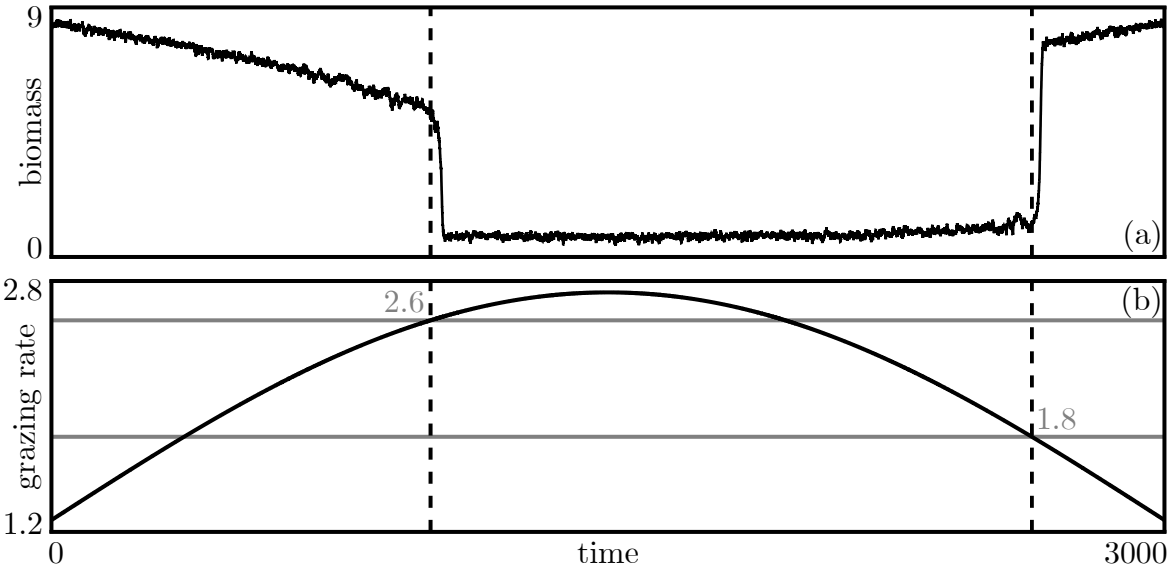


Figure 1.1: A simulation of May's model of ecosystem collapse and recovery: panel (a) shows the system's state as the level of biomass present at a given time; panel (b) shows the input forcing to the system as the grazing rate, also changing over time. Notice how increasingly larger levels of grazing slowly degradate the vegetation biomass in the system until a sudden collapse to a state of almost no cover once the grazing reaches an approximate value of 2.6, indicated by the leftmost dashed black line. The response of the system however does not immediately react in the opposite direction when the grazing rate is brought back to a value below the critical one. Instead, the vegetation cover remains in a state of collapse until the forcing is decreased all the way to approximately 1.8 (corresponding to an approximate 30% reduction from the critical value) in order for the system to recover the previous, healthy state.

herbivores density. This new, unsustainable state is also stable however, meaning that the ecosystem will remain in its collapsed state unless a restoring action is enforced. Interestingly however, simply driving the grazing rate back to pre–breakpoint levels may not suffice to guarantee the reversal of the unhealthy state following the critical transition (as shown in Figure 1.1). This increase in the restoring effort can sometimes be so significant that the tipping point may be de–facto irreversible, leaving the ecosystem in a permanent state of collapse.

Despite the rather severe disruptions caused by such events, critical transitions appear to be ubiquitous across many human and natural systems, with varying degrees of complexity and of substantially different nature. Examples of tipping events leading to catastrophic phenomena can be found in extreme paleoclimatic events [58, 7], ecosystems degradation [69, 57], spikes in neurobiological activity [13], onset of asthma and ventilation defects [85, 38], anthropogenic shut–down of ocean currents

[11, 32], hysteresis in laser systems [47], instabilities in magnetically confined plasma [25] and thermoacustic devices [72], capsizing of marine vessels [82], formation of traffic congenstion [20], financial crashes [84, 54] and civilization collapse [80, 36].

1.1 Historical context

Perhaps unsurprisingly, shortly after May's paper on ecosystems collapse, several attempts in characterizing these catastrophic events in a systematic manner followed. In 1984 Wissel observed that the return time of perturbations towards the stable steady–state increases the closer the system is to the breakpoint [88]. Two years later Wiesenfeld and McNamara realized that, more generically, systems approching bifurcations display amplifications of the spectral properties of their signal due to weakening in the attraction of the favourable steady–state [87]. This *"universal law"* was linked to the concept of resilience in (stochastic) ecological systems by Ives in 1995 which identified an increase in variability of population densities relative to increases in environmental degradations [49].

This characteristic phenomenon, later named *critical slowing down* (CSD) in 1998 [40], usually refers to the lethargic decay of stochastic fluctuations in the neighbour of a local attractor as it approaches a tipping point. In particular, the time for these perturbations to relax back to the stable equilibrium is exponential for an attractor that is far enough from the breakpoint, and algebraic, for the same attractor, close to the breakpoint. The onset of CSD prior to a critical transition has since been heavily exploited by several communities to attempt to forecast catastrophic events their breakpoint.

First, in 2001 Scheffer et al. proposed a model of stochastic resilience in ecosystems in which a gradually changing condition, such as the inflow of toxic chemicals and nutrients, or the loss of species diversity, can be perturbed by seemingly random events such as weather extremes, wild fires or pests outbreaks [79]. The resulting model, corroborated by empirical evidence in lakes and oceans ecosystems, was the first attempt at combining multi–stable, deterministic regimes of long–term behaviour with short–term, stochastic fluctuations in the state of a complex system. This was followed by a similar modeling choice in 2003 by Kleinen and Held, which once again used a stochastic approach to detect treshold switchings in the North–Atlantic thermohaline circulation within both conceptual [53] and intermediate complexity models [46]. Finally, in 2006 Carpenter and Brock used timeseries simulations of a lake model

to forecast its eutrophication, with several decades in advance, by detecting CSD via increases in the signals' standard deviation [16].

These early works in the first half of the 2000s, spurred a significant uptake in the prediction of catastrophic shifts in the stable regimes of ecological and climate systems for the remaining part of the decade; here we cite but a few [59, 67, 18, 30, 43, 9, 23, 43, 15, 17, 29, 39]. By the early 2010s the research question of detecting CSD to forecast tipping points in natural systems had spread quite substantially across many scientific communities up to the point that both early reviews [78, 81] and criticisms [45, 34] started to appear.

1.2 A mathematical framework

All of the early endevours in the prediction of critical transitions so far, while relying on the mathematical characterization of slower return times provided by CSD, did not rigorously quantify the proposed statistical indicators as signals of early–warning. In other words, a lack of mathematical justification, beside few qualitative arguments made thus far, started to, justifiably, raise some concerns and skepticism regarding the range of applicability of this new theory [12, 10, 52, 14].

In 2011, Christian Kuehn proposed a framework based on slow–fast stochastic differential equations (SDEs) for the characterization of critical transitions in complex system and the statistical detection of CSD from their timeseries [55]. The starting point is to consider the following SDE

$$dX_t = f(X_t; \mu)dt + \sigma dW_t, \tag{1.1}$$

where $X_t \in \mathbb{R}^d$ is a collection of $d \geq 1$ random variables representing the state of the system (i.e. its response) at any given time t, $\mu \in \mathbb{R}^p$ is a $p-$dimensional parameter vector representing the external forcing acting on the system (i.e. its input), W_t is a canonical Wiener process of standard increments (i.e. $W_{t+s} - W_t \sim \mathcal{N}(0, s)$), $f : \mathbb{R}^d \to \mathbb{R}^d$ is a sufficiently smooth vector field quantifying the (deterministic) drift of the state variables and $\sigma > 0$ quantifies the magntitude of the noisy fluctuations induced by the diffusion process W_t relative to the drift.

Steady–states of a complex system correspond to equilibria of the drift term in (1.1), that is points $x_* \in \mathbb{R}^d$ in the state space for which the vector field vanishes, i.e. $f(x_*) = 0$. Observable steady–states, that is states for which the complex system is measured in the real–world, are further characterized by the spectral properties of the

linearized vector field f. In particular, an equilibrium $x_+ \in \mathbb{R}^d$ is (linearly) stable if the entirety of the spectrum of the Jacobian of f, evaluated at x_+, is in the complex half–plane with negative real part, i.e.

$$x_+ \quad \text{is stable} \quad \iff \quad \mathfrak{Re}\Big[\mathrm{Spec}\big(Df(x_+)\big)\Big] < 0. \tag{1.2}$$

A stable equilibrium attracts nearby regions of the state space which means that (relatively small) stochastic fluctuations around x_+ will robustly return back to it. Conversely, the Jacobian of f at an (linearly) unstable equilibrium $x_- \in \mathbb{R}^d$ features a spectrum entirely contained in the positive real half of the complex plane, i.e.

$$x_- \quad \text{is unstable} \quad \iff \quad \mathfrak{Re}\Big[\mathrm{Spec}\big(Df(x_-)\big)\Big] > 0. \tag{1.3}$$

To produce critical transitions using (1.1), one must allow for either the existence, stability or both of a stable equilibrium x_+ to be parametrized by the external forcing μ. A prototypical example under this framework is the saddle–node normal form for which $f(x;\mu) = -\mu - x^2$. It is easy to deduce that for every strictly negative value of μ there exist two distinct branches of equilibria of f, namely $x_\pm(\mu) = \pm\sqrt{-\mu}$; the positive branch $x_+(\mu)$ has a negative Jacobian $Df(x_+(\mu)) < 0$, which makes it a stable equilibrium, while the negative branch $x_-(\mu)$ has a positive Jacobian $Df(x_-(\mu)) > 0$, which makes it an unstable equilibrium. At $\mu = 0$ these two branches collide and collapse into a unique equilibrium at the origin $x_+(0) = x_-(0) = 0$ which results in a saddle–node (often called *fold*) bifurcation. Finally, for any strictly positive value of μ no real–valued root of f exists and as such no equilibria are left in the state space $\mathbb{R}$; the system can thus never settle on a steady–state and its fate is to *escape* at infinity (finite time blow-up).

Kuehn noted that, by making the parameter itself time–dependent, one can drive the state of the system across the qualitative change prescribed by the saddle–node bifurcation. Specifically, by setting the deterministic part of (1.1) to be a slow–fast (nonautonomous) drift of the form

$$\begin{aligned} dX_t &= f(X_t, \mu)dt + \sigma dW_t, \\ d\mu &= \varepsilon dt, \end{aligned} \tag{1.4}$$

where $0 < \varepsilon \ll 1$ denotes the timescale separation between the evolution of X_t and μ, an initially stable, observed state $x(\mu_0)$, $\mu_0 < 0$, will inevitably pass through a region

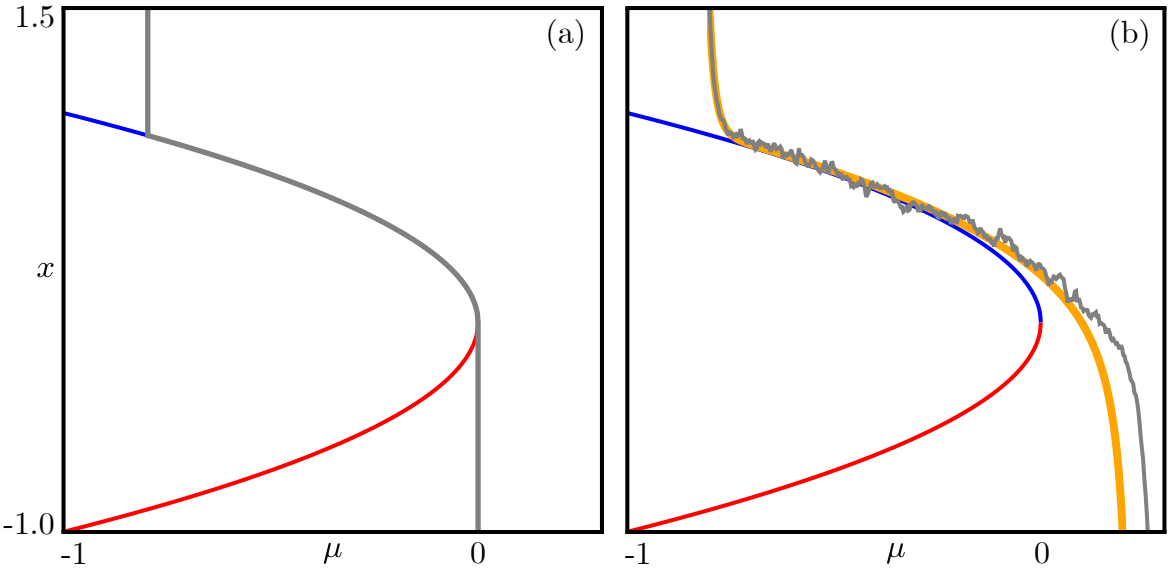


Figure 1.2: Trajectories of a slow–fast system of the form (1.4) associated to a saddle–node normal form $f(x,\mu) = -\mu - x^2$. Panel (a): in the singular limit $\varepsilon \to 0$ a deterministic solution, corresponding to the noiseless case $\sigma = 0$ (grey curve), is decomposed in fast (double arrows) and slow (single arrow) flows; notice that the slow portion of the flow approximates the stable branch $x_+(\mu) = \sqrt{-\mu}$ of the equilibria of f (blue curve). Panel (b): in the nonautonomous regime $\varepsilon = 10^{-1}$ a deterministic solution (orange curve) converges fast to a neighbour of the stable branch and slowly tracks it up to the saddle–node bifurcation after which it escapes at $-\infty$. A sample path associated to a noise level $\sigma = 10^{-2}$ (grey timeseries) follows the same trend of the deterministic solution despite the stochastic fluctuations.

of instability at $\mu = 0$ and eventually be forced to tip to $-\infty$, thereby resulting in a bifurcation–induced critical transition (see Figure 1.2).

To substantiate the statistical detection of CSD from timeseries however, one needs to consider the statistical properties of noisy sample paths rather than just following its deterministic drift. Kuehn thus considered the singular limit $\varepsilon \to 0$ of (1.1) for which a trajectory X_t evolves around the stable equilibrium $x_+(\mu)$, $\mu < 0$ and its fluctuations attain a Boltzmann–like (stationary) probability distribution of the form

$$p(x) \overset{d}{\approx} N \exp\left(\frac{2}{\sigma^2}\int_I f(x;\mu)dx\right), \tag{1.5}$$

with $N > 0$ being a normalization constant of p over a compact set I that contains $x_+(\mu)$. The existence of a stationary (sometimes referred to as *equilibrium*) distribution (1.5) enables the analytical approximation of the central variance of the random

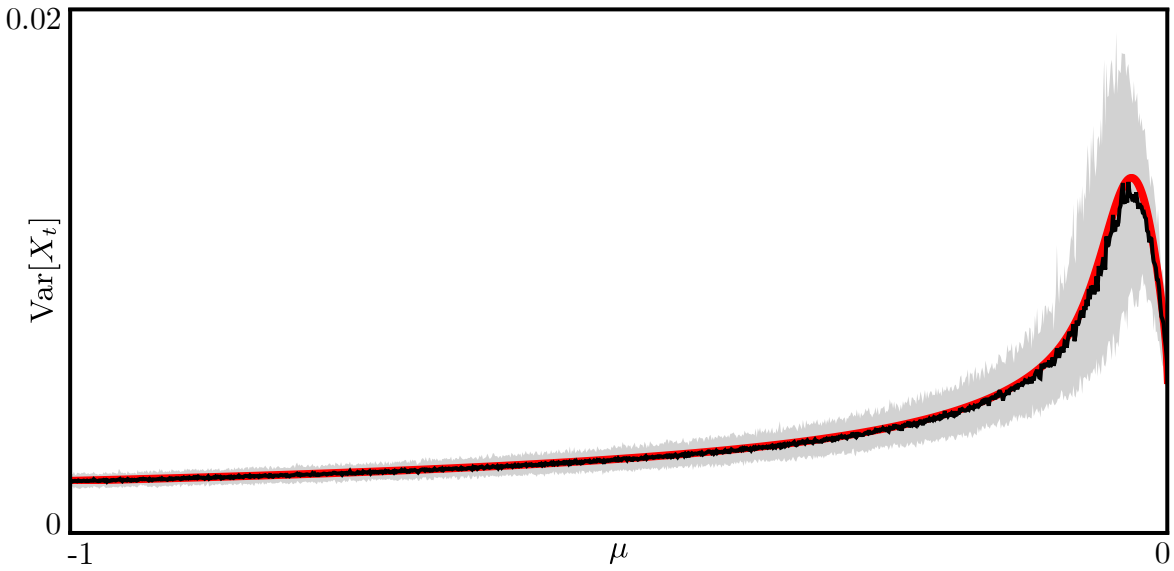


Figure 1.3: Comparison between the variance (1.6) of the equilibrium distribution (1.5) for the saddle–node normal form $f(x;\mu) = -\mu - x^2$ (red curve) and the sample variance of Monte Carlo simulations. At specified parameter values in $[-1, 0)$ an ensemble of 1,000 solutions is computed; for each of these trajectories the sample variance is computed, thus yielding a set of 1,000 measurements. The ensemble mean (black curve) and the $5^{\text{th}} - 95^{\text{th}}$ percentile (gray shaded region) are then extracted.

variable X_t by exploiting its definition

$$\text{Var}[X_t] = \int_I \big(x - \mathbb{E}[x]\big)^2 p(x) dx. \tag{1.6}$$

By considering the case of a saddle–node normal form $f(x;\mu) = -\mu - x^2$, $I = [x_-(\mu), +\infty)$, and plugging (1.5) in (1.6) one can approximate $\text{Var}[X_t]$ by quadrature and thus, in principle, show that an increase in variance of the sample paths does indeed preceed the tipping point. This result is then corroborated by Monte Carlo simulations (see Figure 1.3) in which the variance (1.6) of the equilibrium distribution (1.5) matches the mean sample variance of the solutions of (1.1) with good accuracy.

1.3 Modern developments

In 2012, Ashwin et al. discovered another mechanism that is responsible for generating critical transitions, which is inherently tied to the nonautonomous nature of (1.4). They discovered that when the rate of change ε of the forcing parameter μ is fast enough it would cause the quasi steady–state of a system to escape the basin of

attraction of a stable equilibrium; they named this phenomenon rate–induced tipping [6]. Rate–induced tipping fundamentally differs from the bifurcation–induced counterpart formalized by Kuehn since, in the former case, the system need not to be driven through a local (or global) bifurcation in order for tipping to occur, as later shown in [5, 86]. Ritchie and Sieber later proved that CSD can still be detected for rate–induced tipping via increases in timeseries variance albeit with delay [75]. Rate–induced tipping have since become an active field of research in the modeling of critical transitions, with notable applications recently discovered in ecosystems [69, 57] and ocean currents [3, 60, 19].

Concurrently with these developments, other communities started to research possibilities to extend the theory of CSD–based early–warning signals of tipping points to high–dimensional systems. Specifically, attempts to generalize statistical precursors of critical transitions for spatially–extended systems started to appear in the second half of the 2010s. Early efforts by Kuehn and others on the loss of stability of travelling wave fronts [56] and homogeneous steady–states [42] led the way to later advances in the derivation of early–warning signals for spatial networks which carried over in the early 2020s. Examples of these can be found in [51, 22, 37, 31, 66]. Finally, due to the limitations posed by the collection of large data from spatially extended systems, a question remained on how to optimally observe high–dimensional systems in order to detect CSD. Most recently, works by Lohmann and Gottwald in 2025 [61], and Clarke et al. [24] and Miyauchi et al. [64] in 2026, have started laying the foundations on how low–dimensional observables (i.e. smooth transformations of the state variables) can be extracted from high–dimensional systems in order to anticipate critical transitions through early–warning signals.

1.4 Practical limitations

The mathematical theory of early–warning signals has developed substantially from its early steps, with significant advances been made in the second half of the 2010s and early part of the 2020s. Despite the strong theoretical underpinning provided by the theory, the applicability of these theoretical tools to real–world phenomena remained somewhat limited. In most of the above cited works in fact, the detection of CSD via temporal, spatial or spatio–temporal signatures happens almost exclusively in abstract settings, in which rather strong assumptions on the structure of the system and the data itsels are held. On the contrary however, when the same techniques are applied

data collected from measurements or experiments, rather than simulated, the resulting signals carry significantly less predictive power and robustness across different systems.

In 2015, Dakos et al. noted that the application of CSD detectors to real–world ecosystems is limited by *conceptual*, *operational* and *methodological* constraints [28]. Conceptual limitations arise whenever the abrupt regime changes observed in nature are not due to bifurcation–induced critical transitions but instead due to other mechanisms such as noise– or rate–induced escapes; operationally, the early detection of CSD is only reliable whenever there exist sufficient separation in the temporal and spatial scales governing the evolution of the state of the ecosystem and its external forcing, which is not guaranteed to be true universally [71]; in terms of the methodology itself, the observation protocol with which the data is collected highly impacts the significance and sensitivity of these statistical indicators which thus calls for high quality and large quantity of measurements, which may not be satified due to costs and accessibility constraints [21].

A decade later, Ashwin et al. [4] showed how nonlinear and coupling effects in the dynamics of these complex systems heavily affect the predictive power of early–warning signals (named *early–warning skill*) to detect CSD early enough. They observed that early–warning signals with low skill lead to unreliable predictions in ensemble simulations where the rate of missed allarms (i.e. *false negatives*) and spurious warnings (i.e. *false positives*) severely increases compared to favourable cases. This was recently corroborated in [65] and [35] where high–dimensionality and correlations in the noise structure has been shown to altern the significance of the early–warning signals extracted from observables of these systems.

1.5 Research aim and outline

The above limitations constitute a significant gap between the mathematical abstraction provided by the theory of CSD–based early–warning signals and the real limitations faced by scientists and modelers that seek to apply these methodologies to their data. In this thesis we focused in bridging this gap by providing a more pragmatic approach in the detection of CSD. We leveraged mathematical structures in stochastic and nonautonomous dynamics to build end–to–end data pipelines, that can be applied out–of–the box, to provide direct, interpretable and actionable early–warning signals to scientists, engineers and policymakers. As such, we address two major themes regarding the limitations of applicability of the theory of early–warning signals, which we briefly outline below.

Lack of universal interpretability

At its core, an early–warning signal is a, often statistical, quantity extracted from a monitored observable that somehow provides insights in the proximity of the underlying system to a catastrophic shift in the observed equilibrium. As such, in order for these signals to be actionable, one should be able to interpret such proximity from numerical values attained by the proposed early–warning with reasonable confidence. Rather surprisingly however, the most popular and established detectors of CSD, e.g. the increase in timeseries variance, do not satisfy this requirement. As stated by Dakos et al. in [28] in fact *"one–time measurements of CSD indicators have limited forecasting value"* and only *"once a baseline indicator value is established, then a consistent deviating trend from the baseline could indicate whether a transition is lurking"*. This limitation stems from the fact that traditional CSD–based precursors are not bounded and their numerical value alone does not trivially inform the proximity of a tipping point, rather their trend does. On top of this, systems of different nature, or even the same system under different temporal and spatial scales, do not agree on similar values of variability across time and space, thereby making this limitation a universal feature of the currently used early–warning signals.

To remedy this constraint, we formulate an interpretable early–warning signal that is bounded in $[0, 1]$ for any complex system whose tipping is bifurcation–induced. In particular, values of the proposed quantity close to 0 indicate small likelihood of tipping events to occur in the near future, whereas values close to 1 indicate immediate danger of tipping events to occur in the near future. The proposed early–warning signal is universal in the sense that any system which is slowly driven through a saddle–node tipping, and in the presence or relatively small perturbations, will monotonically increase from 0 to 1, regardless of its nature and its temporal or spatial scales.

Lack of optimal forecasting protocols

Extrapolation methods to estimate the time of tipping of saddle–node types of critical transitions have been proposed for the anticipation of the collapse of the Atlantic Meridional Overturning Circulation by Boers in 2021 [11] and Ditlevsen and Ditlevsen in 2023 [33]. More recently in 2025, Ashwin et al. used the same setup to establish the early–warning skill (or lack thereof) carried by samples of cascading tipping systems [4]. These methods, based on the estimation of the *return* (or *restoring*) *rate* from timeseries data, rely on a set of assumptions that the data must specify in both quality

and quantity. Furthermore, these estimators are negatively biased due to finiteness and non–stationarity of the samples, which previous works do not account for.

To overcome these limitations, we build an alternative pipeline that relaxes some of these assumptions. Such pipeline is entirely data–driven and extends to any saddle–node, bifurcation–induced tipping event perturbed by additive noise. It leverages results in noise–induced escapes and stochastic, slow–fast dynamics to reduce the bias and uncertainty in the local reconstruction of the equilibrium distribution of the noisy fluctuations around a quasi–steady equilibrium. On top of this, we derive optimal conditions for the collection of the data that makes the extrapolation problem feasable and more accurate. The resulting observation protocol thus provides strict and interpretable conditions for scientists and modellers to directly apply forecasting techniques to their timeseries.

Outline

The thesis is organized as follows.

In **Chapter 2** we provide the interdisciplinary mathematical grounding of critical transitions in slow–fast SDEs. The background consists of two main ingredients: stochastic processes, which we review in § 2.1, and dynamical systems, which we review in § 2.2. Starting with basic concepts in both disciplines, these two sections ultimately introduce more advanced topics, such as large deviations and geometric singular perturbation theory, with the aim of introducing the fundamental ideas behind the mathematical modeling of critical transitions and our ability to forecast them.

In **Chapter 3** we tackle the lack of universal interpretability, one of the two research themes of this thesis. In § 3.1, inspired by statistical mechanics and the results provided by large deviation theory, we derive a novel early–warning signal that monotonically increases from 0 to 1 as a generic bifurcation–induced tipping point is approached. In § 3.2, under design assumptions for the underlying system, we introduce a computational pipeline for the estimation of this early–warning signal from timeseries. In § 3.3 we assess the accuracy and sensitivity of the proposed methodology under varying design conditions. Applications that showcase the universality of the early–warning signal to real–world timeseries, and its extraction pipeline, are deferred to Chapter 4.

In **Chapter 4** we address the other research theme of this thesis, which is associated to the problem of forecasting tipping times via extrapolation. In § **??** we ... In § **??** we derive the necessary conditions for an optimal observation protocol to be

employed by the end–users. As for the previous theme, applications of the forecasting methodology and observation protocol are referred to Chapter 4.

In **Chapter 5** we show how both research themes apply to timeseries undergoing bifurcation–induced critical transitions by considering both synthetic and real–world, high–dimensional systems.

Finally, in **Chapter 6** we summarize the main results discussed throughout the thesis and indicate possible future directions for the research to develop.

Chapter 2

Background

In the following, and for the rest of this paper, we will treat functions y that depend both on the state variable x and on time t, i.e. $y = y(x,t)$. To avoid confusion with regards to differentiation we will adopt the following notation

$$\dot{y} := \partial_t\, y(x,t) \quad \text{and} \quad y' := \partial_x\, y(x,t)\,.$$

We will also assume that, for fixed parameters $\mu \in \mathbb{R}^p$ the (frozen) deterministic drift is a conservative vector field, i.e. there exists a scalar potential $V(x;\mu) : \mathbb{R}^d \to \mathbb{R}$ such that $f(x;\mu) = -V'(x;\mu)$. Since we would like to predict B-tipping of saddle-node (SN) type, we assume that the multistability of the system entails that V is a sufficiently smooth function of the state variable x and that it also grows sufficiently fast as $x \to \pm\infty$. The gradient structure of the dynamics allows to map equilibria of the vector field f to stationary points of its potential V. In particular, locally stable equilibria of f correspond to local minima of V while locally unstable equilibria are associated to local maxima of V. This allows for a useful interpretation of the solution of (1.1) as the motion of an overdamped particle governed by the Langevin equation

$$dX_t = -V'(X_t;\mu)dt + \sqrt{2D}dW_t\,, \quad D = \frac{\sigma^2}{2}\,. \tag{2.1}$$

If the process is stationary (i.e. $\varepsilon = 0$) then, for a given configuration of the potential V at parameter value μ, the long term behaviour of a solution of the Langevin equation of motion (2.1) is described by random fluctuations (of intensity $\sqrt{2D}$) around the local minima of V.

2.1 Stochastic processes

2.1.1 Markov processes

A sample path $\mathcal{X} = \{X_n\}_{n=0,\dots,N}$ of (1.1) is an Itô process that is also Markov [70, Ch.7, p. 111]. This results in the joint distribution of the sample to be written in a factorized form of the (marginal) transition probability densities

$$p(\mathcal{X}) = p(X_0) \prod_{n=1}^{N-1} p(X_{n+1}|X_n)\,. \tag{2.2}$$

2.1.2 The Fokker-Planck equation

The FPE is a parabolic partial differential equation in time and space

$$\partial_t p(x,t|\mu) = \partial_x(V'(x;\mu)\,p(x,t|\mu)) + D\,\partial_{xx}^2 p(x,t|\mu)\,. \tag{2.3}$$

Setting its LHS to 0 yields a second-order, homogeneous ordinary differential equation in $p(x|\mu)$ with non-constant coefficients

$$0 = \partial_x\big(V'(x;\mu)\,p(x|\mu)\big) + D\,\partial_{xx}^2 p(x|\mu) = V''(x;\mu)\,p(x|\mu) + V'(x;\mu)\,p'(x|\mu) + D\,p''(x|\mu)\,.$$

The solution of the above (under suitable choices of boundary conditions at $x \to \pm\infty$) yields a Boltzmann-like probability density function [74, Ch. 5, p. 98]

$$p(x|\mu) = \frac{1}{Z(\mu)} \exp\left(-\frac{V(x;\mu)}{D}\right), \tag{2.4}$$

with the normalization constant $Z(\mu)$ playing the role of the partition function of the energy of the system.

2.1.3 The Ornstein–Uhlenbeck process

The OUP reads

$$dX_t = \omega(\tilde{x} - X_t)dt + \sqrt{2D}dW_t. \tag{2.5}$$

The stationary density of the OUP is [41, Ch. 3, p. 74]

$$p(x|\mu) = \sqrt{\frac{\omega}{\pi D}} \exp\left(-\frac{\omega(\tilde{x}-x)^2}{D}\right), \tag{2.6}$$

The saddle–node normal form with additive noise is

$$dX_t = (-\mu - X_t^2)dt + \sigma dW_t\,. \tag{2.7}$$

It can be approximated by an Ornstein–Uhlenbeck process of the form

$$dX_t = J(x_+ - X_t)dt + \sqrt{2D}dW_t, \tag{2.8}$$

where the shift corresponds to the stable branch (1.2) of the equilibria of $f(x;\mu) = -\mu - x^2$, i.e. $x_+(\mu) = \sqrt{-\mu}$, and the linear coefficient corresponds to the Jacobian of f evaluated at such branch, i.e. $J(\mu) \coloneqq Df(x_+(\mu)) = -2\sqrt{-\mu}$.

2.1.4 Large deviations theory

In statistical mechanics, Kramer's formula quantifies the rate at which particles escape across potential barriers when subject to Langevin's equation of motion (2.1). Kramer's formula is sometimes expressed as the reciprocal of the escape time, usually referred to as the (mean) first exit time. Different derivations of the formula are available in the literature; we thus redirect the readers to [8] for a concise overview. It must be noted that, while limitations on Kramer's law vailidity exist (see [8]), our assumptions of stationary, 1−dimensional dynamics subject to small, additive white noise fall within the applicability cases.

Given a double-well potential V with desirable local minimum (stable equilibrium) $x = a$ and local maximum (unstable equilibrium) $x = b$, the rate at which particles, starting at $x = a$ and subject to white noise of intensity $\sqrt{2D}$, will escape over the saddle $x = b$ is given by

$$R_K = \frac{\sqrt{|V''(x_-)|V''(x_+)}}{2\pi} \exp\left(-\frac{\Delta V}{D}\right), \tag{2.9}$$

where $\Delta V_{a\to b} = V(b) - V(a)$ quantifies the potential barrier that the Langevin overdamped particle has to overcome in order to escape, and $V''(a), V''(b)$ quantify the local curvature of V around the well and the hilltop respectively. The large deviation principle applied to the saddle–node normal form (2.7) gives

$$R(\mu) = \exp\big(-\Delta V(\mu)\big) = \dots \tag{2.10}$$

2.2 Dynamical systems

Introduce the autonomous saddle-node normal form and an example of coordinate transformations for the saddle-node bifurcation theorem. Extend it to the dynamic case with the notion of pullback attractors.

2.2.1 Geometric singular perturbation theory

Main results of Fenichel theory.

Chapter 3

An interpretable early-warning signal

In this chapter we address the first of the two research themes of the thesis; the lack of universal interpretability of early–warning signals across systems of different scale and nature. We detect CSD in timeseries by exploiting a large deviation principle of the rare, noise–induced escapes, of solutions out of the basin of attraction of stable equilibria; in practice, the proposed measure is a modified version of Kramer's escape formula (2.9). After showing the universal scaling of the new early–warning signal, we construct a pipeline that computationally extract such quantity from the timeseries by means of parameter estimation of the confining potential.

3.1 Formulation of the early–warning signal

Among the many EWS that have been proposed in the literature, the increase in variance has been the most popular. Its success primarly stems from a relative simplicity in computation and a relative robustness in detecting CSD. The major drawback of using the increase in variance is, however, its lack of interpretability. As discussed in the introduction, numerical values of the variance of a sample fail to provide useful interpretation if one does not have access to a baseline value for the system when at a safe distance from the bifurcation–induced tipping.

In this section we show that a modified version of Kramer's escape formula solves this issue by providing an interpretable early–warning signal. Let us consider a saddle–node normal formal perturbed by small, additive noise (2.7). As discussed in § 2.1.4, for any $\mu \ll 0$, the large deviation principle provided by Kramer's formula (2.9) holds. In this regime, the two wells of the potential are well separated and thus the local minimum (stable equilibrium) $x = x_+$ is much deeper than the local maximum

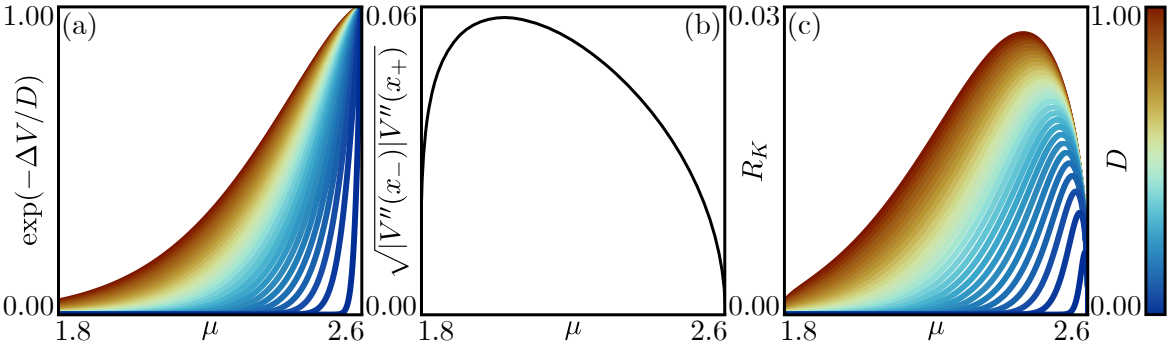


Figure 3.1: Comparison of the the noise–dependent exponential term of the escape early–warning signal $R(\mu)$ (panel (a)) and Kramer's escape rate R_K (panel (c)) at different values of D (colorbar on the right). The prefactor term of Kramer's escape rate (panel (b)) is independent on D and thus shown by a single black curve.

(unstable equilibrium) $x = x_-$. In other words, far from the bifurcation the potential barrier $\Delta V = V(x_-) - V(x_+) \gg 0$. Conversely, as the saddle–node bifurcation is approached, the stable equilibrium $x = x_+$ and the unstable equilibrium $x = x_-$ of the system become closer, resulting in the potential barrier reducing, which implies that $\Delta V \to 0$.

Based on this scaling, we can easily conclude that the large deviation principle (2.10) can be a detector of CSD since

$$R(\mu) = \exp(-\Delta V(\mu)) \approx \begin{cases} 0\,, & \mu \ll 0\,, \\ 1\,, & \mu \to 0\,. \end{cases} \tag{3.1}$$

Notice that (3.1) is obtained from (2.9) by removing the curvature-dependent prefactor $\sqrt{|V''(x_-)|V''(x_+)}/2\pi$ and setting $D = 1$ in the argument of the exponential. Removing the prefactor guarantees that the modified formulation (3.1) increases monotonically as $\mu \to 0$. Conversely, the prefactor in (2.9) makes Kramer's formula initially increase away from zero, before reaching a maximum and then decreasing back to zero at the bifurcation $\mu = 0$ (see Figure 3.1(c)). On the other hand, setting $D = 1$ guarantees that the modified formulation is not late in detecting the proximity of the tipping point whenever the underlying noisy fluctuations are small $D \to 0$ (see Figure 3.1(a)). Furthermore, the true value of D is generally not known a–priori and, while estimation is possible, it will inflate the dimensionality of the problem without providing significant advantages to the underlying detection of CSD.

The boundedness of (3.1) in $(0, 1]$ solves the interpretability problem we outlined above. As opposed to the variance (and higher-order moments), a numerical value

of the modified escape formula provides direct interpretation of the likelihood of a tipping event without the need for a baseline value to refer to. If a system is far from a dangerous transition (i.e. $\mu \ll 0 \implies \Delta V \gg 0$), then (3.1) will be close to 0; conversely, a state that is approaching said transition (i.e. $\mu \to 0 \implies \Delta V \to 0$) results in values of the same quantity to be close to 1.

In Figure 3.2 we depict the monotonic increase of (3.1) from 0 to 1 for May's model of ecosystem collapse we introduced in Chapter 1. The region of bistability (depicted by the gray area in Figure 3.2 (a,c)) is bounded by two saddle–node bifurcations at μ_1 and $\mu_2 > \mu_1$. The earlier saddle–node bifurcation at $\mu = \mu_1$ involves the (unhealthy) stable equilibrium $x = x_{+,2}$ and the unstable saddle $x = x_-$ while the later saddle–node bifurcation at $\mu = \mu_2$ involves the (healthy) stable equilibrium $x = x_{+,1}$ and the same saddle x_-. This bifurcation sequence is reflected in the shape of the potential V shown at a number of parameter slices in Figure 3.2 (b1-b5), wherein the local maximum x_- and the local minima $x_{+,1/2}$ correspond to the unstable and stable equilibria in Figure 3.2 (a), respectively. Outside the shaded region (i.e. $\mu < \mu_1$ or $\mu > \mu_2$, e.g. Figure 3.2 (b1,b2,b5)) only one stable equilibrium exists (either $x_{+,1}$ or $x_{+,2}$), the potential V has one global minimum and no local maxima and thus the modified escape early–warning (3.1) is undefined. Conversely, inside the shaded region (i.e. $\mu_1 < \mu < \mu_2$, e.g. Figure 3.2 (b3,b4)) the potential V has two minima $x_{+,1/2}$ and one local maximum x_-, meaning that the modified escape early–warning signal (3.1) can be computed (e.g. Figure 3.2 (c)).

Notice that $R(\mu)$ increases monotonically with μ, with its minimum being at $\mu = \mu_1$ and its maximum being at $\mu = \mu_2$. Therefore, by identifing the subset of the parameter space $\mathbb{R}$ such that ΔV has range $[0, +\infty)$, we associate to $R(\mu)$ the intepretation of the likelihood of a tipping event to occur.

3.2 Estimation of the early–warning signal

Figure 3.2 shows that $R(\mu)$ can be used as an interpretable early–warning signal whenever the potential V of a gradient system is known which, in most cases, it is not. As mentioned above, the popularity of canonical early–warning signals, such as the increase in timeseries variance, is in part due to their simplicity in calculating them from available data. In order for our proposed, interpretable early–warning signal to be applicable in the real–world we must be able to compute it from timeseries data without relying on prior knowledge of the system that generates it.

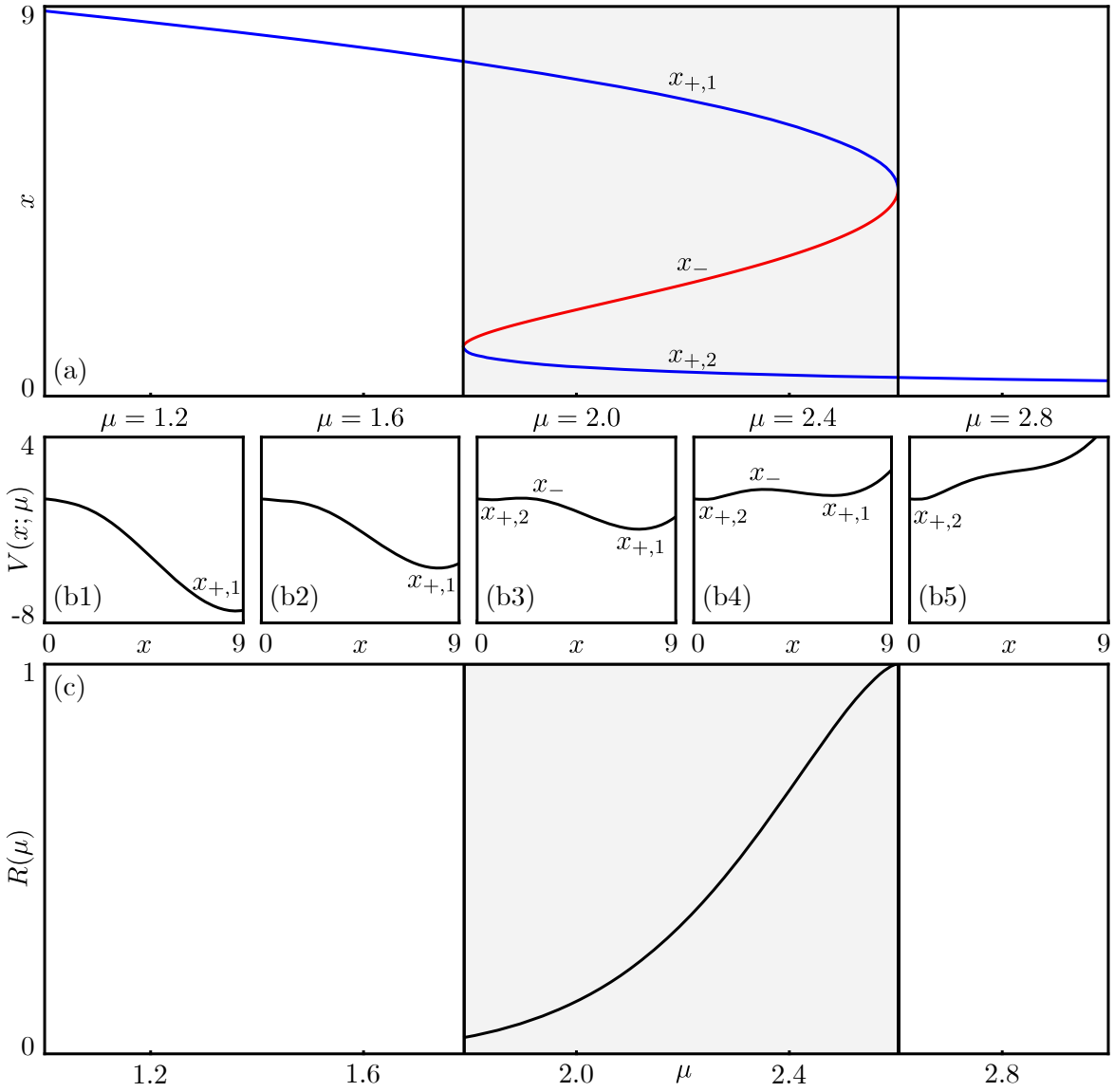


Figure 3.2: (Panel (a)) Bifurcation diagram May's model of ecosystem collapse with the state x (vegetation biomass) degrading as the forcing parameter μ (grazing rate) is increased. Branches of stable equilibria ($x = x_{+,1}$ and $x = +, 2$) are in blue while the unstable branch ($x = x_-$) is in red; the region of bistability, bounded by two saddle–node bifurcations bifurcations, is shaded in gray. (Panels (b1-b5)) Different configurations of the potential $V(x;\mu) = -\frac{x^2}{2} + \frac{x^3}{30} + \mu\big(x - \arctan(x)\big)$ at different parameter values (ticks of the horizontal axes of panels (a) and (c)). (Panel (c)) Modified escape early–warning signal (3.1) computed out of the potential $V(x;\mu)$ between the stable state $x_{+,1}$ and the saddle x_- within the bistability region.

Since the modified escape rate (3.1) depends on the bifurcation parameter μ through the potential barrier ΔV between the observed steady-state x_+ and an un-

known unstable equilibrium x_-, our aim for this section is to estimate such quantity from a single timeseries. To do so we must be able to approximate the local shape of the potential $V(x;\mu)$ between x_+ and x_-. The inference of the polynomial coefficients of the potential V is, at its core, a parameter estimation problem of the deterministic drift of a SDE given a sequence of its realizations. This is a well-established problem in statistical inference [77, Ch. 11, p. 234]; in the following we briefly formulate the setup for such problem and redirect interested readers to [68, 83, 26] for comprehensive reviews.

3.2.1 Problem formulation

A standard technique to infer the parameters of a stochastic model of the form (1.1), using its solutions, is maximum likelihood estimation (MLE). Suppose our sample $\mathcal{X} = \{X_n\}_{n=0,1,\dots,N}$ is made of $N+1$ observations of the sample path solving (1.1) from time $t = 0$ to time $t = T$. Suppose also that each observation is sampled at homogeneous frequency of time intervals $\Delta t = T/N$. As established in § 2.1.1 this Itô process is a Markov process which means that the transition probability density to a future state X_{t+1}, at any time t, solely depends on the current state X_t rather than the entire history $\mathcal{X}$. As a result of this property, the joint distribution of the sample factorises into the transition probability densities as in (2.2), which allows us to define the likelihood of $\mathcal{X}$ to be

$$\mathcal{L}(\theta|\mathcal{X}) := p(X_0|\theta) \prod_{n=1}^{N-1} p(X_{n+1}|X_n,\theta)\,, \tag{3.2}$$

parametrized by $\theta \in \mathbb{R}^p$. By maximizing (3.2) one finds the most likely set of parameters θ^*, for a family of transition densities $p(x|\theta)$, to have generated the observations $\mathcal{X} = \{X_n\}_{n=0,\dots,N}$. Notably one reformulates this problem in the minimization of the inverse log-likelihood instead

$$\ell(\theta|\mathcal{X}) = -\log \mathcal{L}(\theta|\mathcal{X}) \approx -\sum_{n=1}^{N-1} \log p(X_{n+1}|X_n,\theta)\,, \tag{3.3}$$

as this does not change the stationary point in the search space

$$\theta^*(\mathcal{X}) = \arg\max_{\theta\in\mathbb{R}^p} \mathcal{L}(\theta|\mathcal{X}) = \arg\min_{\theta\in\mathbb{R}^p} \ell(\theta|\mathcal{X})\,. \tag{3.4}$$

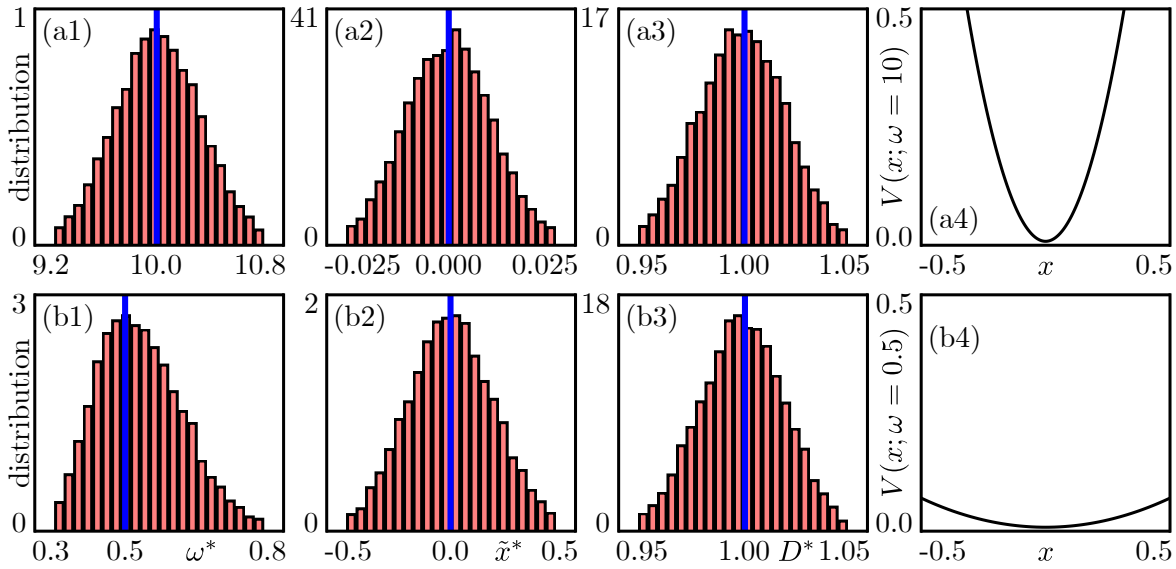


Figure 3.3: Distribution of the MLE of the parameters $\theta^* = (\omega^*, \tilde{x}^*, D^*)$ from an ensemble of 10,000 sample paths solving (2.5) for two different configurations of the potential $V(x) = \omega(\tilde{x} - x)^2$: a deep, strongly confining well with $\omega = 10$ (panel (a4)) and; a shallow well with $\omega = 0.5$ (panel (b4)). In each of the panels (a1-a3) and (b1-b3) the true value of the parameter is reported (blue line).

Notice that in (3.3) we approximate the inverse log-likelihhod by ignoring the conditional probability of the deterministic initial condition X_0 as this term is negligible compared to the sum of the log-likelihoods of the stochastic observations. The optimization problem (3.3) requires the knowledge of the transition density $p(X_{n+1}|X_n, \theta)$, which, for a SDE of the form (1.1), is goverened by the Fokker-Plank equation (2.3).

3.2.2 Time-invariant approximation

In the specific case of slow-fast systems of the form (1.4), the computation of the transition density, despite not being tractable analytically, can be approximated. Specifically, in the singular limit $\varepsilon \to 0$ the (linear) ramp of the forcing parameter is much slower than the timescale of the evolution for the state variable. This means that the time-varying solution of the Fokker–Planck equation (2.3) can be reasonably approximated with a time-invariant density $p(x|\theta) \approx p(x, t|\theta)$ by neglecting the (partial) time derivative $\partial_t p \approx 0$. As established in § 2.1.2, thies yields the Boltzmann-like equilibrium distribution (2.4) and thus, in principle, one can use it as the transition density in the optimization problem (3.4).

As an example of the applicability of this workflow, let us consider the simplified case of an Ornstein-Uhlenbeck process (2.5) with linear coefficient $\omega > 0$ and shift

$\tilde{x} > 0$. By denoting $\theta = (\omega, \tilde{x}, D)$ to be our parameter vector, we setup a MLE problem in which the (limiting) Gaussian distribtion of the process is given by (2.6) with quadratic potential $V(x) = \omega(\tilde{x} - x)^2$. For large enough values of the linear coefficient ω, the solution of this MLE problem has been shown to be asymptotically normal [1], as demonstrated in Figure 3.3 (a1-a3). When $\omega \to 0$ however, the distribution of ω^* becomes skewed towards larger values (see Figure 3.3 (b1)). On top of this, the standard deviation of the distribution of the estimated shift $\tilde{x}^*$ increases (see Figure 3.3 (b2)), yielding an overall less accurate estimation. Recall from § 2.1.3, that we can approximate the equilibrium sample path of a saddle–node normal form (2.7) using an Ornstein–Uhlenbeck process of the form (2.8), with the mapping $\omega \mapsto J(\mu) = -2\sqrt{-\mu}$ and $\tilde{x} \mapsto x_+(\mu) = \sqrt{-\mu}$. We can thus conclude that, since $\omega \to 0$ as $\mu \to 0$, the MLE of a system, slowly driven through a saddle–node tipping, using the stationary distribution of the Ornstein-Uhlenbeck process decreases in accuracy as the bifurcation is approached.

3.2.3 Euler-Maruyama approximation

A desirable property of an early–warning signal is that its prediction becomes more accurate and reliable when the monitored system is in a proximity of the critical transition. Conversly in fact, an early–warning signal that becomes less accurate when the system is close to tipping will systematically fail to detect CSD, which has arguably more catastrophic implications than providing spurious warnings. Estimating the likelihood of tipping events using the Ornstein–Uhlenbeck process, as described above, achieves the opposite of this, as its accuracy decreases the closer the sample path is to the saddle–node bifurcation. In this section we overcome such limitation by using an alternative approach to MLE.

As anticipated at the beginning of the section, we want to infer the likelihood of a bifurcation–induced tipping by approximating the local shape of the confining potential so that the modified escape rate $R(\mu)$ can be computed. The minimum model that we can fit to a timeseries, that is assumed to be a sample path of a Langevin equation (2.1), is that of a cubic polynomial $V(x;\theta) = \sum_{k=1}^{3} \theta_k x^k$ with real coefficients $\theta = (\theta_1, \theta_2, \theta_3)$. While different choices are available, a cubic polynomial is minimal in the sense that it allows for the presence of one local minimum x_+ and one local maximum x_- (from which $\Delta V = V(x_-) - V(x_+)$ can be computed) without inflating the dimensionality of the search space. In other words, we fit a cubic polynomial to a timeseries because that is the model of a saddle–node normal form, which is ultimately the modeling goal for replicating (and forecasting) critical transitions via CSD.

A number of computational methods to approximate the stationary density $p(x)$ are available [73, 50, 1]. Conversely, other techniques have been devised that do not rely on the numerical approximation of the transition density such as the generalized method of moments [44], Markov chain Monte-Carlo algorithms [76, 2] and quasi–maximum likelihood regressions [27, 89]. Here we focus on the latter approach and in particular we describe one quasi–maximum likelihood technique that reduces the MLE problem into an ordinary least–squares regression.

As usual, we assume our sample path $\mathcal{X} = \{X_0, \dots, X_N\}$ to be a solution of (1.1) under the stationarity restriction. Such sample path can be approximated [62, § 8.4] by the following process

$$X_{n+1} = X_n - V'(X_n;\theta)\Delta t + \sqrt{2D\Delta t}\,\zeta\,, \quad \zeta \sim \mathcal{N}(0,1)\,, \tag{3.5}$$

which corresponds to the Euler-Maruyama time–stepping scheme, a generalization to stochastic processes of the Euler method for ordinary differential equations. We rearrange Eq. (3.5) to define the first-order time increments

$$Y_n := \frac{X_{n+1} - X_n}{\Delta t} = -V'(X_n;\theta) + \sqrt{\frac{2D}{\Delta t}}\,\zeta = -V'(X_n;\theta) + \xi, \tag{3.6}$$

where $\xi \sim \mathcal{N}(0, 2D/\Delta t)$. Notice how the iterated map (3.6) defines a regression to a model V' of some data $\{X_n\}_{n=0,\dots,N-1}$ given a set of N observations $\{Y_n\}_{n=0,\dots,N-1}$ perturbed by some (Gaussian) noise ξ. This sets up a parameter estimation problem in which one wishes to find an optimal parametrization θ^* of the potential such that the residual between the model predictions $V'(X_n;\theta^*)$ and the observations Y_n is minimized in some sense. As anticipated above, we formulate an ordinary least–squares problem by interpreting the noise ξ to be unresolved fluctuation errors in the observed data. We thus minimize the residual vector

$$r(\theta) = \big(Y_n + V'(X_n;\theta)\big)_{n=1,\dots,N-1}\,, \tag{3.7}$$

between the observations Y_n and the fitting model $V'(X_n,\theta)$ in the least–squares sense. That is, we solve the minimization problem

$$\theta^* = \underset{\theta\in\mathbb{R}^3}{\arg\min}\, ||r(\theta)||_2\,, \tag{3.8}$$

where $||\cdot||_2$ denotes the L_2-norm. Notice that by fitting the minimal cubic model $V(x;\theta)$, $\theta = (\theta_1, \theta_2, \theta_3)$, the residual (3.7) is linear in θ. This allows us to rewrite

Algorithm 1: Computation the modified escape rate R from timeseries.

Input: sample $\mathcal{X} = \{X_0, X_1, \ldots, X_N\}$ (a single timeseries)
Output: estimated likelihood of tipping R

1. Compute quasi-stationary residuals $\overline{\mathcal{X}}$ using a detrending technique.
2. Assemble the ordinary least–squares problem (3.6)
   for $(X_n, X_{n+1}) \in \overline{\mathcal{X}}$ **do**
   $$y_n = X_{n+1} - X_n\,,$$
   $$A_n = \left(X_n,\ X_n^2,\ X_n^3\right).$$
   end
3. Solve the normal equation (3.10)
   $$\theta^* = (\theta_1^*, \theta_2^*, \theta_3^*) = (A^T A)^{-1} A^T y\,.$$
4. Compute the local minimum and local maximum of $V^*(x) = \sum_{k=1}^{3} \theta_k^* x^k$
   $$x_{\min} = +\frac{1}{3\theta_3^*}\left(\sqrt{(\theta_2^*)^2 - 3\theta_1^*\theta_3^*} - \theta_2^*\right),$$
   $$x_{\max} = -\frac{1}{3\theta_3^*}\left(\sqrt{(\theta_2^*)^2 - 3\theta_1^*\theta_3^*} + \theta_2^*\right).$$
5. Compute the potential barrier
   $$\Delta V^* = V^*(x_{\max}) - V^*(x_{\min})\,.$$
6. Extract the modified escape early–warning signal using Eq. (3.1)
   $$R = \exp(-\Delta V^*)\,.$$

(3.8) as a minimization problem on the quadratic form

$$\theta^* = \underset{\theta \in \mathbb{R}^3}{\arg\min}\, ||A\theta - y||_2\,, \tag{3.9}$$

where $y \in \mathbb{R}^N$ denotes the observation data vector (i.e. $y_j = Y_j$), $A \in \mathbb{R}^{N\times 3}$ is the model matrix evaluated on the sample $\mathcal{X}$ (i.e. $A_{jk} = X_j^k$) and $A\theta - y = r(\theta) \neq 0$. The unique solution of (3.9) satisfies the so-called normal equation

$$\theta^* = (A^T A)^{-1} A^T y = A^+ y\,, \tag{3.10}$$

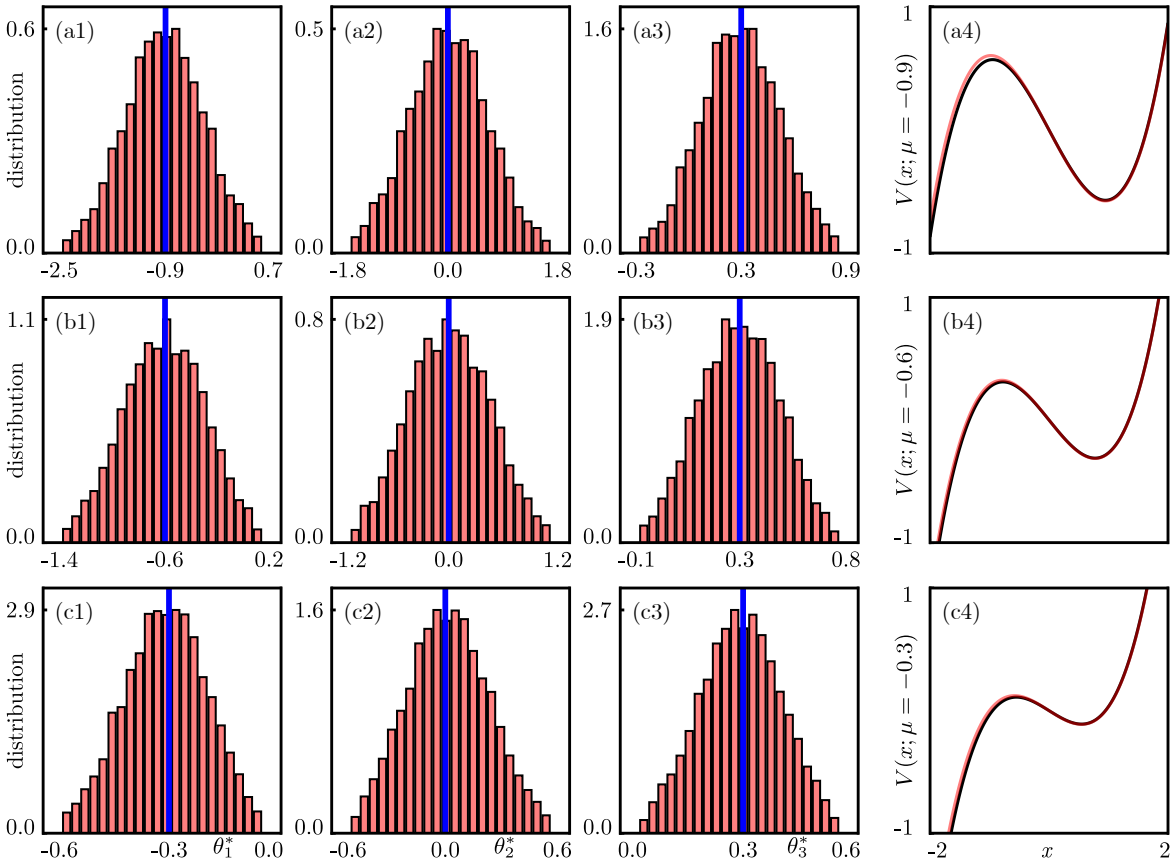


Figure 3.4: Distribution of the solutions $\theta^* = (\theta_1^*, \theta_2^*, \theta_3^*)$ of the ordinary least–squares problem (3.10) for an ensemble of 5,000 sample paths of a saddle–node normal form $f(x;\mu) = -\mu - x^2$. The problem is solved for different configurations of the potential $V(x;\mu) = \mu x + \frac{x^3}{3}$; at $\mu = -0.9$ (panels (a1-a4)), $\mu = -0.6$ (b1-b4) and $\mu = -0.3$ (c1-c4). The value of the polynomial coefficients of the true potential are reported in each of the panels (a1-a3), (b1-b3) and (c1-c3) (blue line). The ensemble mean values of the least–squares solutions $\hat{\theta}^* = (\hat{\theta}_1^*, \hat{\theta}_2^*, \hat{\theta}_3^*)$ are used to plot an ensemble mean reconstruction (red curve in (a4-c4)) of the potential $\hat{V}^*(x) := V(x;\hat{\theta}^*) = \sum_{k=1}^{3} \hat{\theta}_k^* x^k$ and compared with the ground truth (black curve in (a4-c4)).

where $A^T A$ is a Gram matrix and $A^+ \in \mathbb{R}^{3\times N}$ is the Moore-Penrose inverse of A. With the unique solution $\theta^* = (\theta_1^*, \theta_2^*, \theta_3^*)$ one can thus estimate the early–warning signal R of the sample by its definition (3.1). We report the main steps of this procedure in Algorithm 1. The solution of the normal equation (3.10) is also asymptotically normal as shown in Figure 3.4.

Notice that, contrary to the MLE of the Ornstein–Uhlenbeck process described in § 3.2.2, the numerical accuracy of (3.10) improves as the saddle–node bifurcation is approached. This is shown in Figure 3.4 by a reduction in the standard deviation of

the distribution (histograms) of the least–squares solutions θ^* as $\mu \to 0$. We remark that this property is generic (i.e. it is observed for gradient systems other than the saddle–node normal form); we argue that is is a desirable consequence of CSD. As the bifurcation approaches, $J = Df(x_+) \to 0$ as $\mu \to 0$, meaning that the stable attractor weakens the pull towards the noisy perturbations; this results in the state of the system to explore larger portions of the basin of attraction of such x_+. As a consequence, the realizations $\mathcal{X} = \{X_0, \dots, X_N\}$ of the gradient system will capture the local shape of the confining potential in an increasingly better fashion as $\mu \to 0$, which translates in more accurate solutions of the regression (3.10).

3.3 Performance analysis

The computational pipeline in Algorithm 1 allows for the modified escape early–warning signal R to be estimated from slowly varying timeseries following the local reconstruction of the confining potential. However, the statistical significance and uncertainties associated with such method can be heavily affected by the timescale separation $0 < \varepsilon \ll 1$, noise intensity $0 < \sigma \ll \sqrt{\varepsilon}$, sampling frequency $\Delta T > 0$ and sample lengths $N \gg 0$. In the following section quantify such uncertanities in varying regimes of these parameters. Specifically, we perform statistical analysis of the relevant quantities involved in the estimation of the modified escape early–warning signal following Algorithm 1. As a representative example, we consider a one-dimensional, slow–fast system of the form

$$\begin{aligned} dX_t &= \big(-\mu(t) + 4x - 4x^3\big)dt + \sigma dW_t\,, \\ \mu(t) &= \mu_0 + \varepsilon t\,, \end{aligned} \tag{3.11}$$

where the bistable drift $f(x, \mu) = -\mu + 4x - 4x^3$ features a saddle–node bifurcation at $\mu = \mu_c \approx 1.53$.

3.3.1 Robustness of the residuals

One of the assumptions of the MLE of the modified escape early–warning signal is that the underlying distribution of the sample path is stationary. However, solutions $\mathcal{X} = \{X_0, \dots, X_N\}$ of slowly ramped systems, such as (3.11), only satisfy this condition in the singular limit $\varepsilon \to 0$ and for long enough time horizons $N \to \infty$. For strictly positive rates $\varepsilon > 0$ the deterministic drift f is nonautonomous and thus the Fokker–Planck equation (2.3) does not admit a Boltzmann-like stationary solution

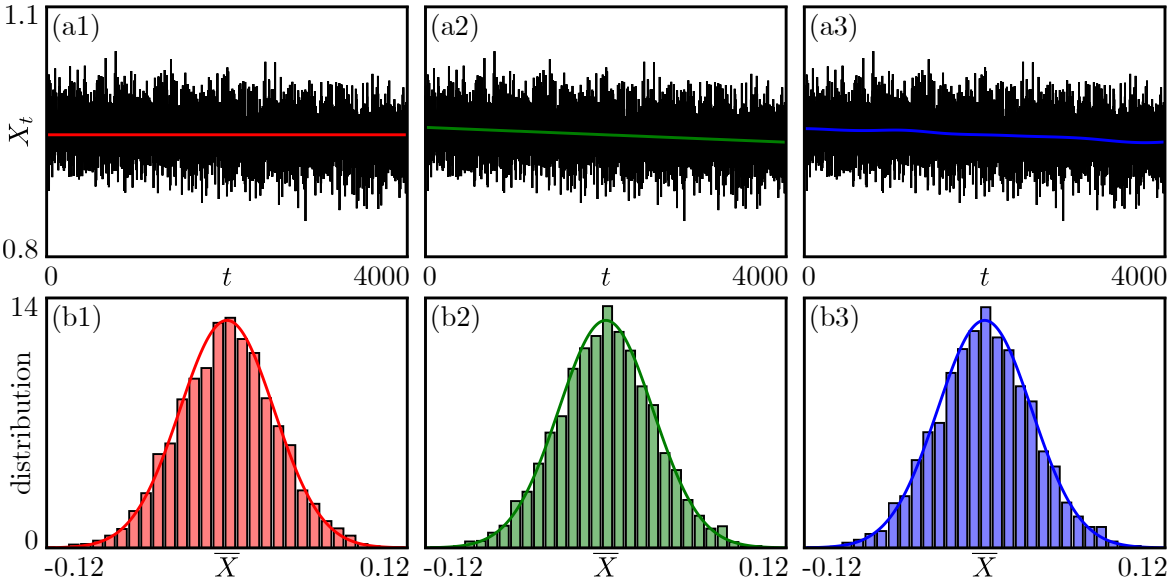


Figure 3.5: Comparison of different detrending techniques for a solution $\mathcal{X}$ of (3.11) with $\mu_0 = 0$, $\varepsilon = 5 \cdot 10^{-4}$ and $\sigma = 0.1$; the simulation stops at time $T = 4{,}000$ at which the bifurcation parameter attains the value $\mu(T) = 0.2$. Panels (a1-a3): the estimated trends of the ramped timeseries (black curves) are computed using the sample mean (red curve), linear fit (green curve) and empirical mode decomposition (blue). Panels (b1-b3): comparison between the stationary density of the Ornstein–Uhlenbeck process (2.6) approximating $\mathcal{X}$ (colored curves) and the empirical distribution of the residuals obtained by sample mean (red histogram), linear fit (green histogram) and empirical mode decomposition (blue histogram).

(1.5). Nonetheless, by considering slow-fast regimes for which $0 < \varepsilon \ll 1$, we can obtain a quasi-stationary sample $\overline{\mathcal{X}} = \{\overline{X}_1, \ldots, \overline{X}_N\}$ by removing the underlying trend in the ramped sample path $\mathcal{X}$. The detrending of non-stationary timeseries (see step 1. in Algorithm 1) is a fundamental pre-processing stage in the derivation and analysis of early–warning signals as the quality of the extracted quasi-stationary residuals can affect the performace of the estimation.

One method to do that is the empirical mode decomposition [48] which is a data–driven technique that extracts the underlying trend of a timeseries through a sifting procedure. Specifically, the empirical mode decomposition iteratively decomposes a timeseries in so-called intrinsic mode functions, each capturing the different scales of fluctuations in the timeseries. The trend of a sample path $\mathcal{X}$ thus corresponds to the intrinsic mode function associated to the largest scale of fluctuation (see [48] for details) and the residuals $\overline{\mathcal{X}}$ are obtained by removing the leading one from the sample path. A desirable property of this technique is that it does not require fine calibrations

Table 3.1: Relative errors statistics (3.12) of an ensemble of 1,000 residuals obtained by different detrending techniques.

	Centering			Linear fit			Empirical mode decomposition		
	min	mean	max	min	mean	max	min	mean	max
E_1	$5 \cdot 10^{-19}$	$4 \cdot 10^{-17}$	$4 \cdot 10^{-16}$	$5 \cdot 10^{-19}$	$1 \cdot 10^{-16}$	$6 \cdot 10^{-16}$	$2 \cdot 10^{-6}$	$3 \cdot 10^{-4}$	$2 \cdot 10^{-3}$
E_2	0.02	0.03	0.05	0.03	0.04	0.05	0.03	0.04	0.05
E_3	$2 \cdot 10^{-5}$	0.03	0.16	$6 \cdot 10^{-5}$	0.03	0.19	$1 \cdot 10^{-4}$	0.03	0.18

as the intrinsic mode functions are extractred through the envelopes of the timeseries. This makes the application of the empirical mode decomposition particularly suitable in the computation of quasi-stationary residuals $\overline{\mathcal{X}}$ as it readily extends to univariate timeseries with nonlinear trends. Similar data-driven techniques, often used in applications, are centering (i.e. removal of the sample mean from the timeseries) and linear fit (i.e. removal of the slope from the timeseries).

To verify the quality of the residuals obtained by these detrending methods we simulate an ensemble of 1,000 sample paths of (3.11) ramped between $\mu_0 = 0.0$ and $\mu(T) = 0.2$, $T = 4{,}000$ with $\varepsilon = 5 \cdot 10^{-4}$. Concurrently, an Ornstein–Uhlenbeck process of the form (2.5) is constructed, approximating the slowly ramped sample paths around the drifting quasi–steady equilibrium $x_+(\mu)$ with linear coefficient $\omega = Df(x_+)$ and $\tilde{x} = x_+(\mu)$ being evaluated at the end of the ramp, i.e. $\mu(T) = 0.2$. Figure 3.5 shows a representative example of this experiment where a sample path of the simulated ensemble (black curves in panels (a1-a3)) is detrended by centering (red curve, panel (a1)), linear fit (green curve, panel (a2)) and empirical mode decomposition (blue curve, panel (a3)). The distribution of the quasi-stationary residuals (histograms in panels (b1-b3)) are then compared to the stationary distribution of the OUP (see Eq. (2.6)). To quantify the deviation of the residuals from the stationary assumption we compute the relative error between the residuals' statistics $\overline{q}_k$ and the central moment q_k of the Ornstein–Uhlenbeck process

$$E_k := \frac{|\overline{q}_k - q_k|}{|q_k|}\,, \tag{3.12}$$

where $k = 1, 2, 3$ denote the first (mean), second (variance) and third (skewness) moment of the distribution. The results of this experiment, compiled in Table 3.1, show that, within the slow–fast hypotheses $0 < \varepsilon \ll 1$, the three detrending techniques largely preserve the stationary assumption with average relative errors well within 10%.

3.3.2 Accuracy and sensitivity

The statistical properties of the least–squares solutions are prescribed by the normally distributed uncertainties $\xi \sim \mathcal{N}(0, \sigma^2/\Delta t)$ in the first–order increments as per (3.6). As a result, the accuracy of $\theta^* \in \mathbb{R}^3$, and hence the MLE of the modified escape early–warning signal R, is reflected by the specific choices of additive noise intensity σ and sampling frequency Δt. Furthermore, it is reasonable to expect that shorter (i.e. smaller N) and faster (i.e. larger ε) timeseries would produce significantly different residuals and thus hinder the stationary assumption discussed above.

We thus investigate the sensitivity of our pipeline when the above constraints (long and slow timeseries with small noise and large sampling frequency) are relaxed. Specifically, we test the accuracy of Algorithm 1 in two different settings:

- by varying the timeseries length N and sampling frequency Δt in the singular limit $\varepsilon \to 0$ and in a regime of small noise $\sigma \to 0$ and;
- by varying the parameter's rate of change ε and noise level σ with fixed timeseries length N and sampling frequency Δt.

In both cases we run an ensemble simulation of 1,000 sample paths of (3.11) and evaluate the accuracy of the proposed method by computing the a–posteriori relative error (3.12) between the estimated modified escape early–warning signal and the ground truth (3.1). In order to assess whether the convergence in mean persists in different regimes, we compare the relative errors at different parameter values as the system is driven closer to tipping.

The results of this experiment are depicted in Figure 3.6. In the stationary case (panels (a1-a3)) we observe that the estimated escape early–warning signal is more accurate for longer timeseries $N \to 10^4$ and larger sampling frequencies $\Delta t \to 10^{-1}$. This is not surprising considering that the uncertainty in the ordinary least–squares regression is inversely proportional to Δt. Furthermore, we notice that the increase in accuracy close to tipping is preserved for varying N and Δt as the mean relative error for the system far from the saddle–node bifurcation $\mu = 1.1$ (panel (a1)) is overall larger than the one for the system closer to the saddle–node bifurcation $\mu = 1.3$ (panel (a3)). In the slow-fast regime (panels (b1-b3)) we keep the timeseries length $N = 10^4$ and sampling frequency $\Delta t = 10^{-1}$ fixed while we generate solutions of (3.11) at varying rates ε and noise intensities σ. For each ε we keep $\mu(T)$ fixed and adjust μ_0 so that each sample path contains exactly $N = 10^4$ realizations. In particular we prescribe $\mu(T) \in \{1.1, 1.2, 1.3\}$ in panels (b1), (b2) and (b3) respectively; these

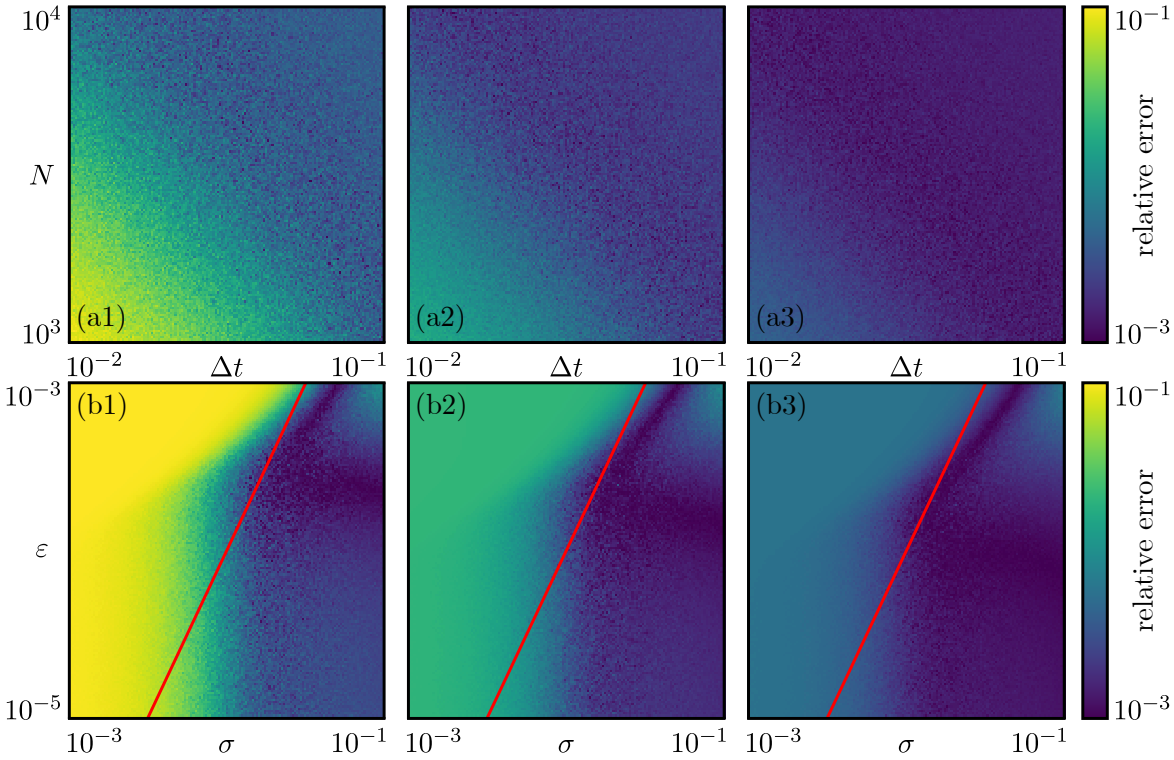


Figure 3.6: Sensitivity of Algorithm 1 under different regimes. Each pixel in the heatmap corresponds to the ensemble mean relative error of 1,000 sample paths of (3.11). Panels (a1-a3): in the stationary case ($\varepsilon = 0$) and with fixed small noise ($\sigma = 0.05$) longer timeseries, sampled at larger frequencies, yield a more accurate estimation of the modified escape early–warning signal. Panels (b1-b3): in the slow-fast regime, for fixed timeseries length $N = 10{,}000$ and sampling frequency $\Delta t = 0.1$, the accuracy of the estimated modified escape early–warning signal depends, non-trivially, from the level of additive noise in the system with smaller noise seemingly yielding less accurate estimations. The red curve represents the $\sigma = \sqrt{\varepsilon}$ threshold.

parameter values are then used to compute the ground truth of the modified escape early–warning signal using (3.1). Intuitively, we observe that faster rates $\varepsilon \to 10^{-3}$ yield overall less accurate estimations due to the residuals deviating further from the stationary assumption (as discussed in the previous section). On the other hand, the estimations become less accurate for smaller noise levels $\sigma \to 10^{-3}$; while Kramer's law holds in regimes of small noise, the least–squares estimation of the confining potential requires larger excursions around the stable equilibrium to better infer the local shape of the basin of attraction. In particular, we notice that the non-trivial decay of the relative error shows a complex relationship modelling the interplay of ε and σ in different regimes, meaning that slower and less noisy timeseries do not necessarily yield better estimates. This emphasizes the fact that the MLE of the modified escape

early–warning signal is sensitive to perturbations in forced systems and it shows that the threshold $\sigma = \sqrt{\varepsilon}$ (red curves in panels (b1-b3)) separating regimes of slow passage through the bifurcation from those of noise–induced escapes, provides an actionable guideline for applications.

3.3.3 Statistical significance

The convergence in mean of the estimated modified escape early–warning signal guarantees accurate warnings when an ensemble of sample paths is available. In real-world applications however, this is rarely the case as access to i.i.d. samples of a given system is limited by physical and technical constraints. We therefore investigate how significant the upward trend for the estimated modified escape early–warning signal is for individual realizations of (3.11). In particular, we are interested in establishing whether the estimations from single sample paths can lead to missed allarms (i.e. false negatives) and/or spurious warnings (i.e. false positives).

To do so we perform an ensemble simulation of 1,000 sample paths ramped from $\mu_0 = -1.6$ to $\mu(T) = \mu_c$ with $\varepsilon = 1 \cdot 10^{-5}$. We then use Algorithm 1 to estimate the modified escape early–warning signal across a sliding window containing $N = 10^4$ realizations of each sample path. We repeat the same procedure for stationary simulations of (3.11) at parameter values $\mu_1 = 0$ (i.e. at a safe distance from tipping) and $\mu_2 = 1$ (closer to tipping but still at a safe distance). We report the results of this analysis in Figure 3.7. The sample paths (black curves in panels (a1-a3)) are computed across a time horizon $T = 3.2 \cdot 10^5$. For the slowly ramped case (panel (a1)), the timeseries will all tip, from the drifting stable equilibrium x_+ (blue curves), across the unstable threshold x_- (red curve), within the prescribed time horizon T; this provides a sample of positive tipping events. For the stationary cases (panels (a2-a3)) the solutions will stay bounded around a small region of the stable equilibrium (blue line) and they will not tip across the unstable threshold (red line) within the time horizon; this provides a sample of negative tipping events. In the ramped case, the estimated ensemble mean of the modified escape early–warning signal (red curve in panel (b1)) overestimates the proximity to tipping given by the ground truth (black curve) partly due to less amount of data available; notice that the familiar convergence in mean, as $\mu \to \mu_c$, implies that the overestimation bias vanishes close to tipping. Furthermore, while the interquartile range (green shaded region) shows a reasonable bounded region for the estimated signal, the outliers in the ensemble (blue shaded region) deviate significantly from the mean. This feature of the MLE of the modified

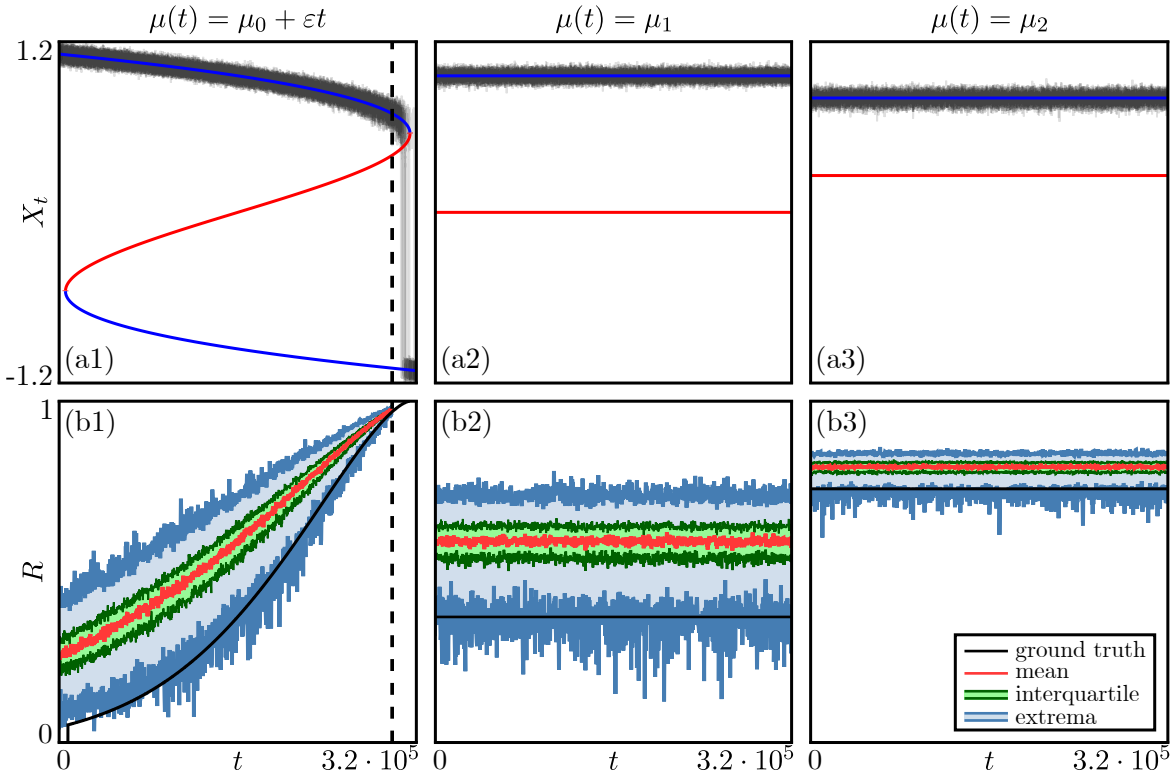


Figure 3.7: Time-varying estimations of the modified escape early–warning signal for an enseble of 1,000 slowly ramped (panels (a1)-(b1)) and stationary (panels (a2-a3)-(b2-b3)) solutions of (3.11). In the ramped case the parameter varies between $\mu_0 = -1.6$ to $\mu(T) = 1.6$ with rate $\varepsilon = 1 \cdot 10^{-5}$. The sample paths, shown by the black curves in panel (a1), follow the stable slow manifold (blue curve) before tipping across the unstable branch (red curve). The earliest tipping in the ensemble is indicated by the vertical dashed line. In the stationary cases, simulated at parameter values $\mu = 0$ (panel (a2)) and $\mu = 1$ (panel (a3)), the system is in the singular limit $\varepsilon \to 0$ and thus the solutions, shown by the black curves in panels (a2-a3), are confined around the stable equilibrium (blue line) and do not tip within the simulated time horizon. In all cases, a rolling window of 10% the total length of the sample paths, is used to estimate R via Algorithm 1 (panels (b1-b3)). The ensemble mean early–warning, shown by the red curves, is compared with the ground truth of the frozen system, depicted by the black curves, alongside its descriptive statistics (see legend in panel (b3)).

escape early–warning signal is also observed in the stationary cases with the deviations reducing as the system is driven closer to tipping.

We conclude that, the overestimation bias in the ensemble mean, and the large deviations of the outliers from such mean can lead to spurious warnings when a single trajectory is considered. However, considering that the interquartile range provides a considerably smaller region around the mean, and that such deviations vanish as

$\mu \to \mu_c$, we can also conclude that the overall trends of the estimations become more significant as the system approaches the saddle–node bifurcation, which minimizes the occurence of missed alarms from single trajectories.

Chapter 4

Forecasting tipping times by extrapolation

Chapter 5

Applications

Chapter 6

Conclusions

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